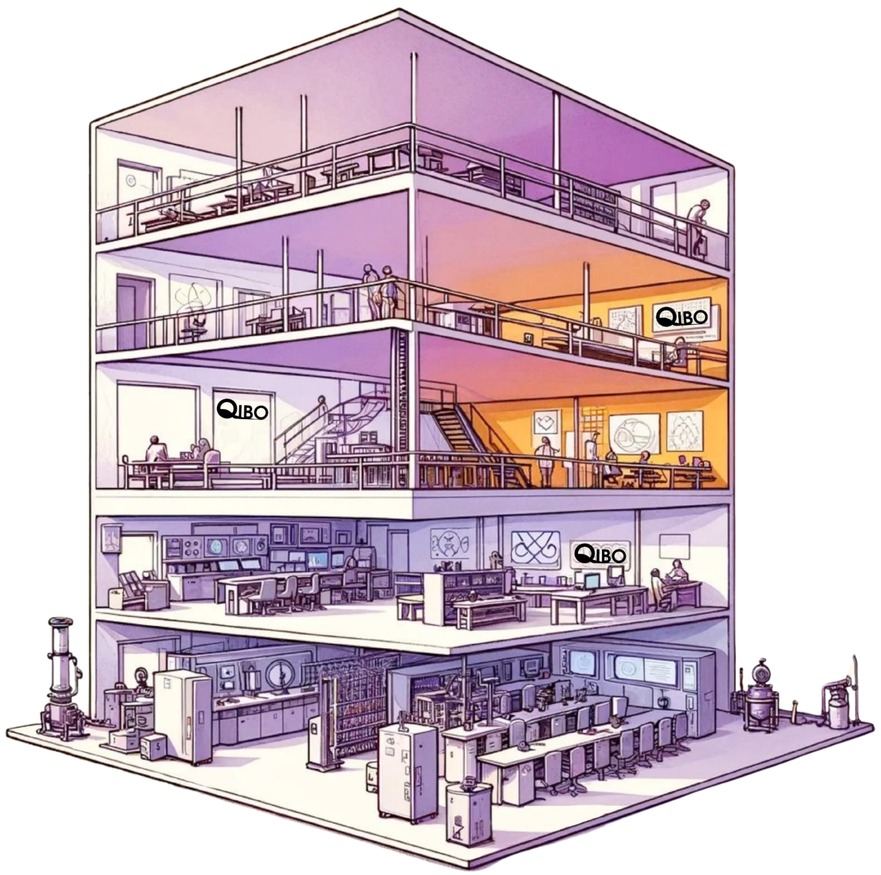
The Qibo Ecosystem: Integrating Advanced QML with Real-Time Hardware Control

Prof. Dr. Stefano Carrazza
University of Milan and INFN Milan, Italy

stefano.carrazza@unimi.it
https://qibo.science



UNIVERSITÀ DEGLI STUDI DI MILANO

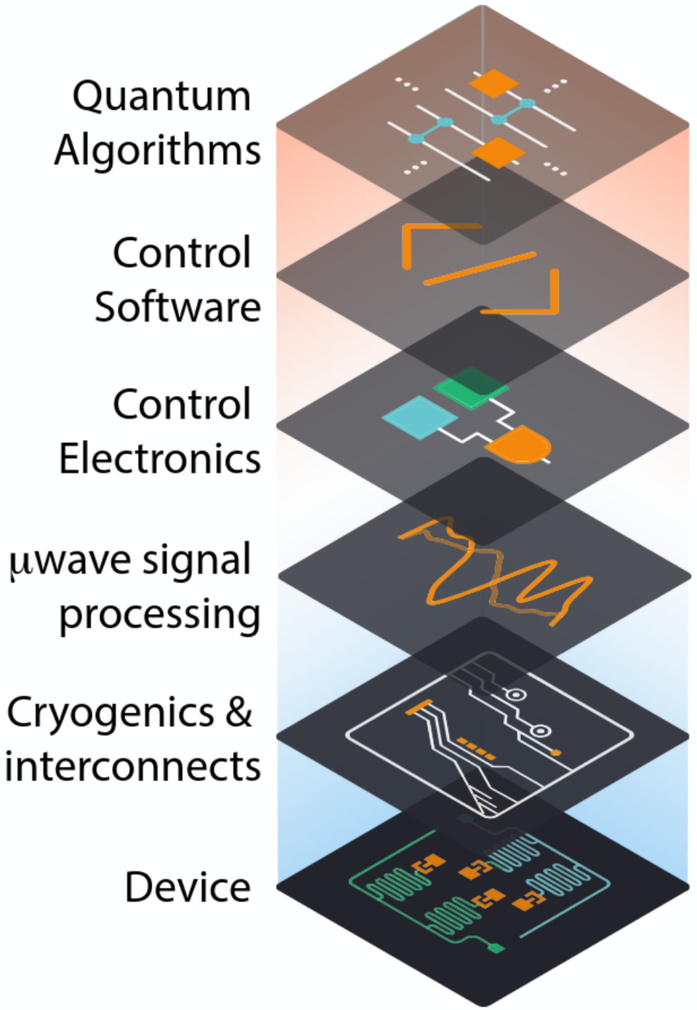
INFN Istituto Nazionale di Fisica Nucleare

NQSTI National Quantum Science and Technology Institute

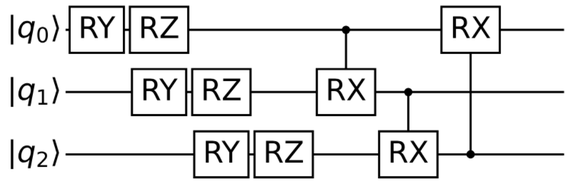
QIBO

A Full-Stack Open Source Quantum Middleware

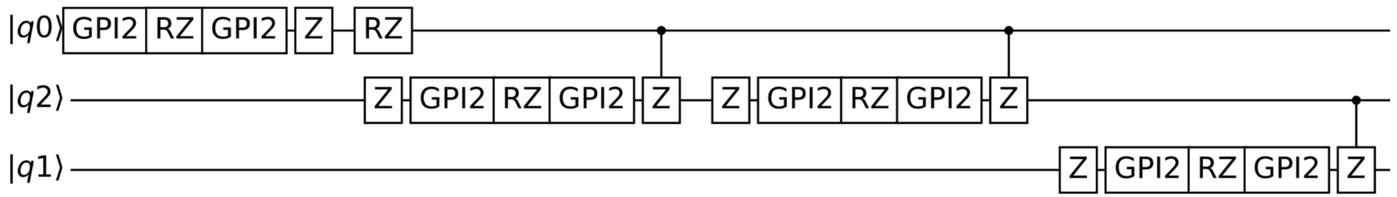
A Multi-Layer Problem



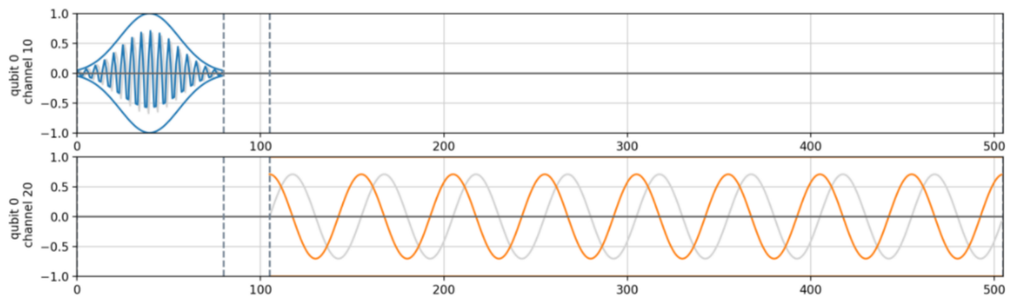
- **High Level**: device-agnostic model definition



- **Intermediate Level**: transpilation to the device-specific topology and native gates set



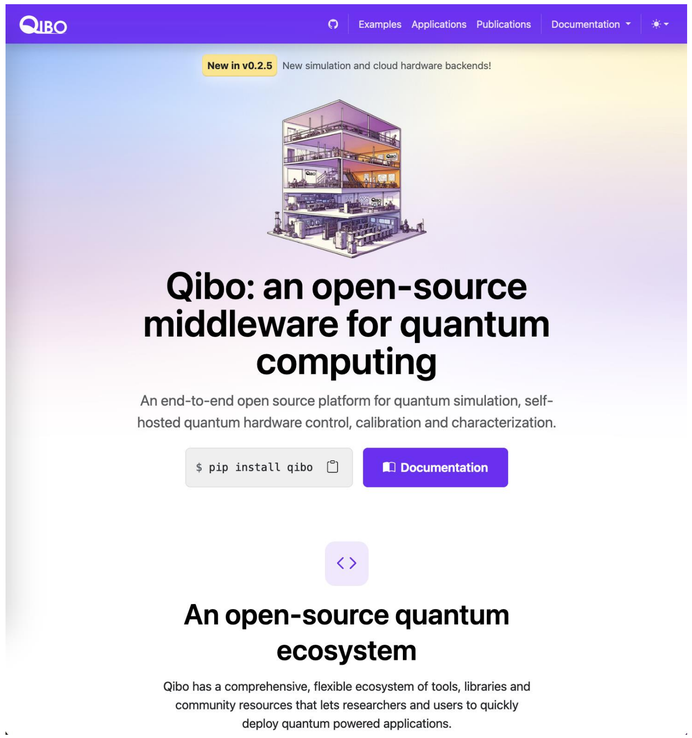
- **Low Level**: gate compilation into microwave pulses



Quantum Middleware

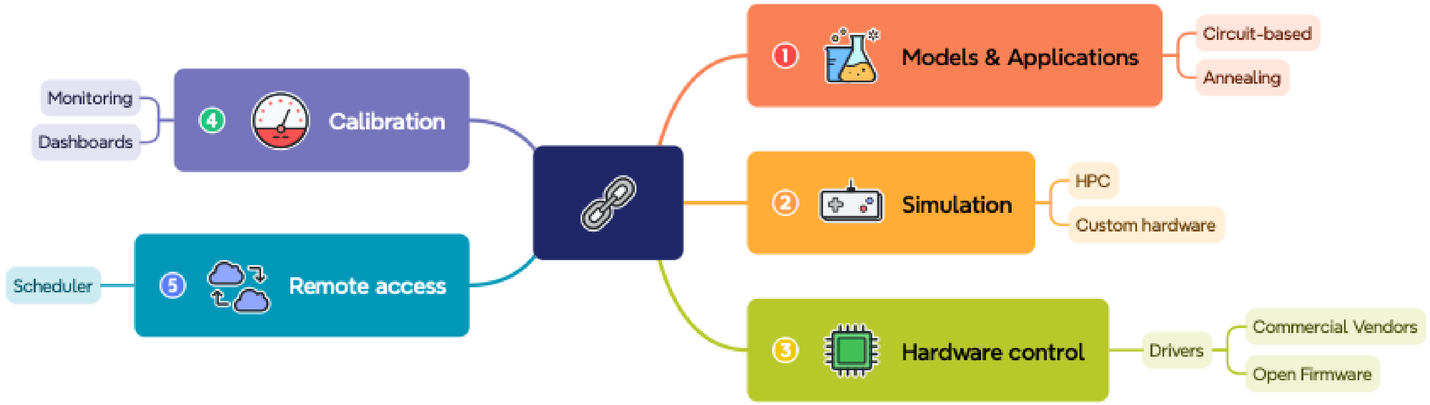
Qibo, is a modular, open-source framework for quantum computing, control and calibration.

Qibo collaboration, arXiv:2009.01845

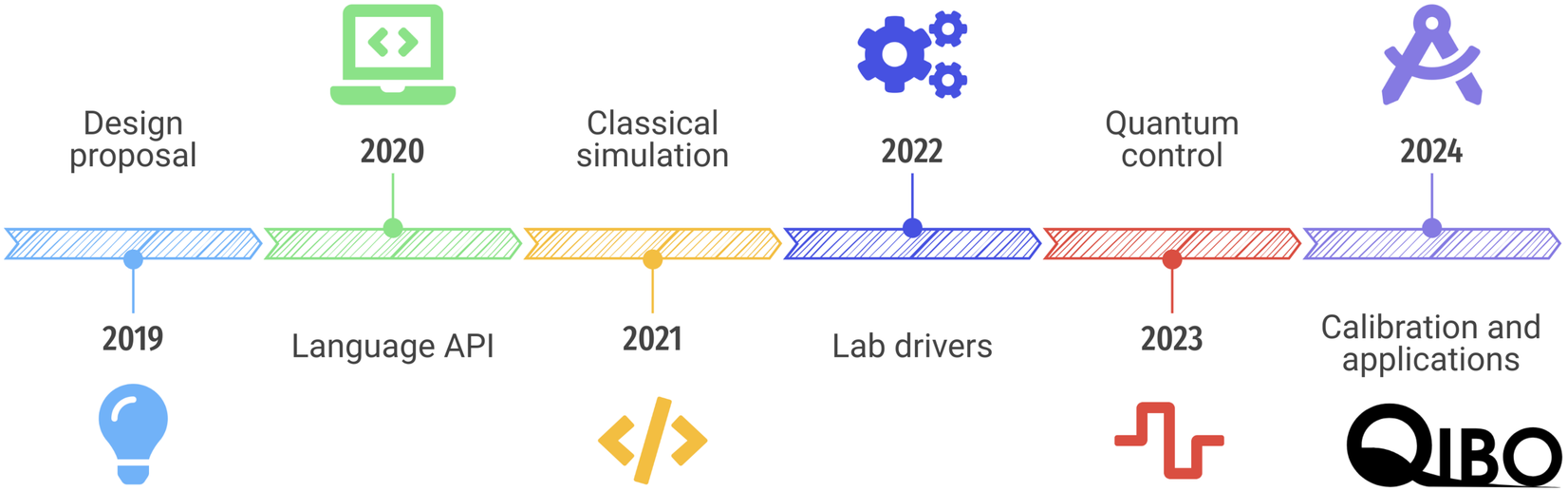


https://qibo.science

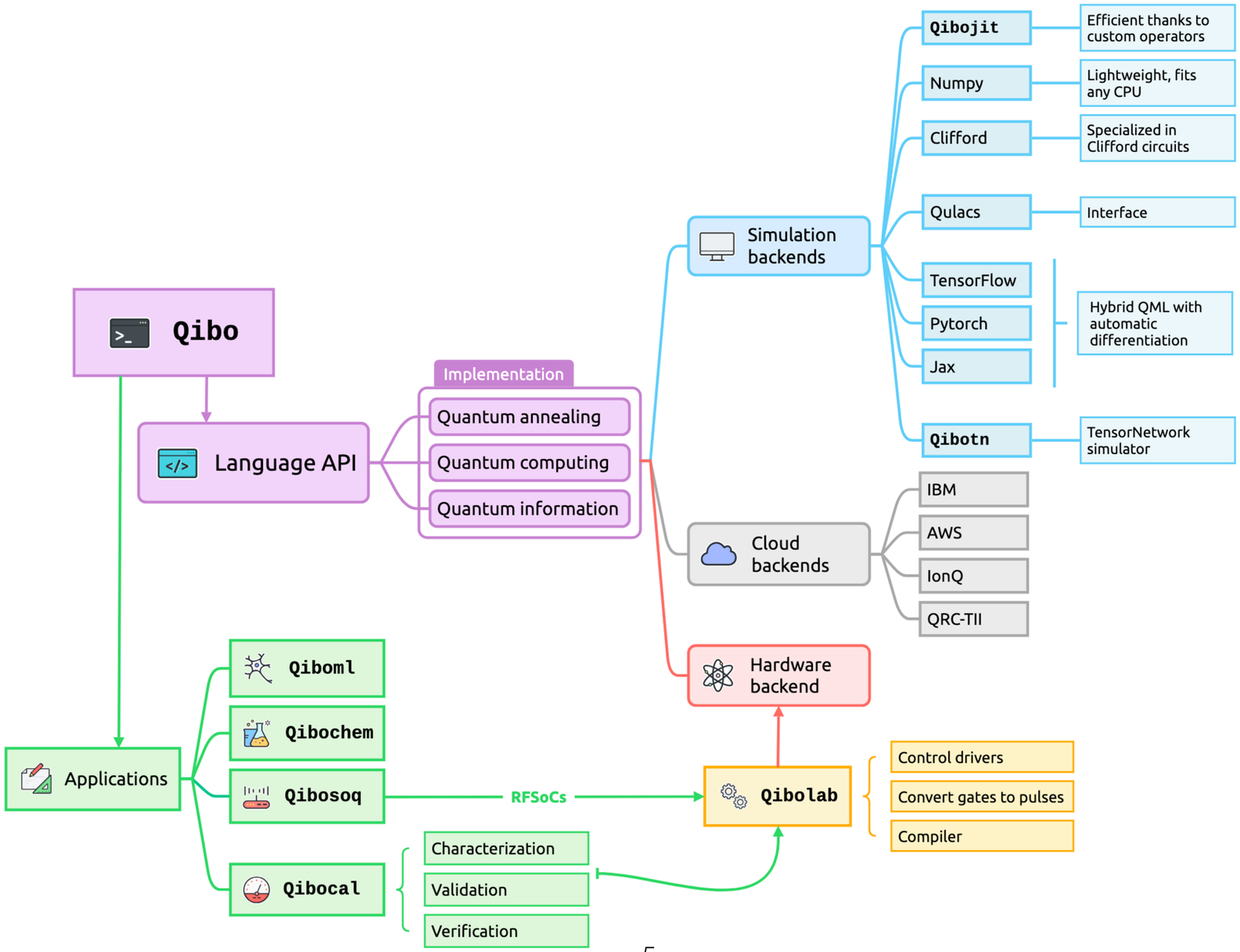
Middleware challenges



Qibo development timeline

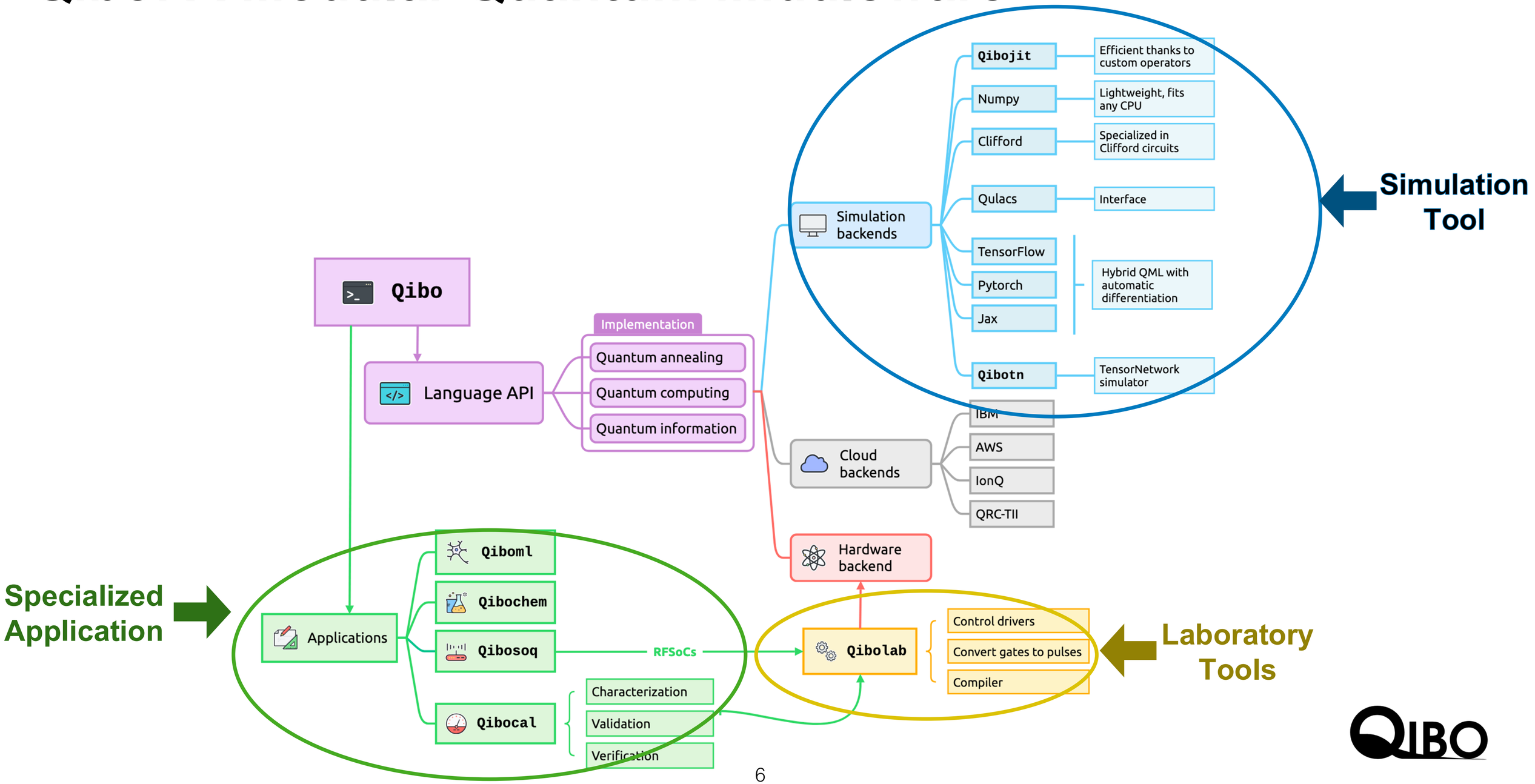


Qibo: A Modular Quantum Middleware



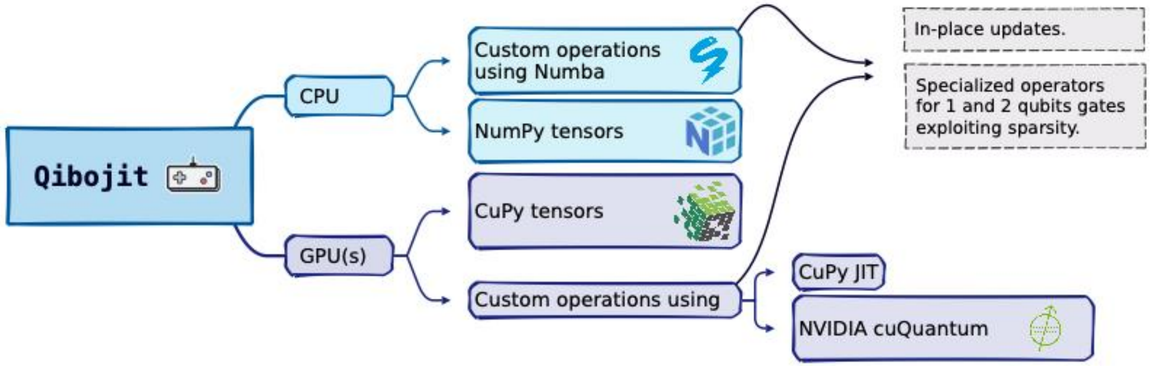
Details:

Qibo: A Modular Quantum Middleware

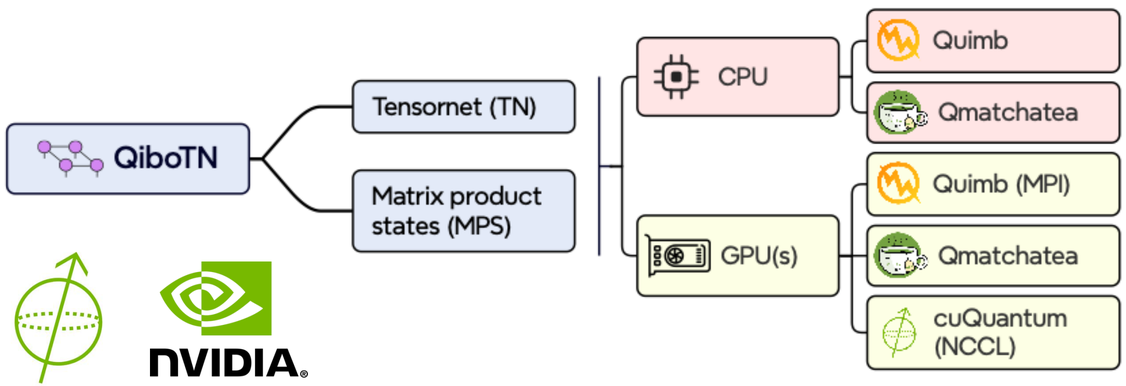


Quantum Simulation

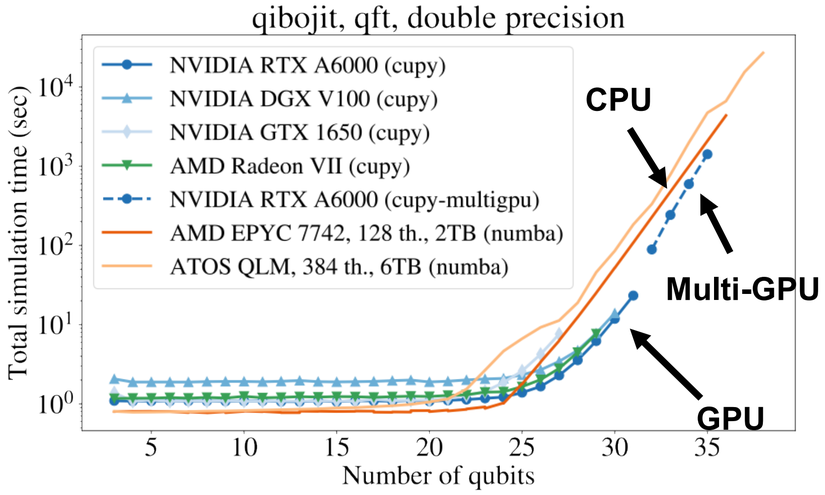
State-Vector Simulation



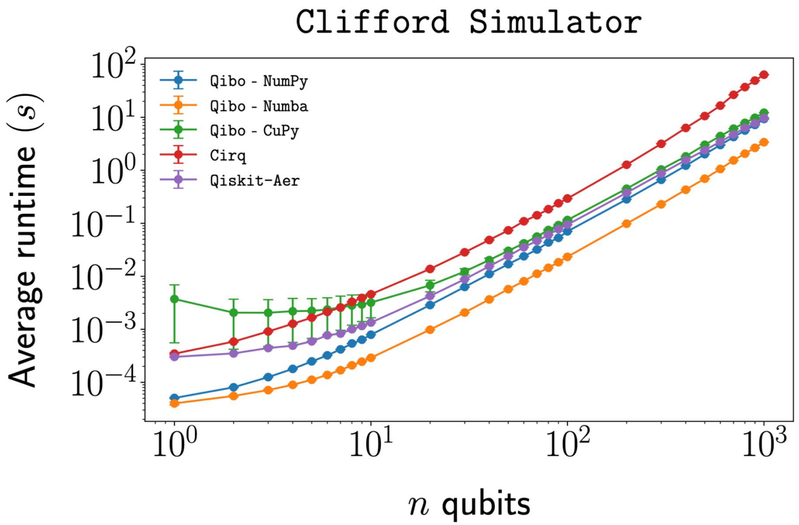
Tensor Network Simulation



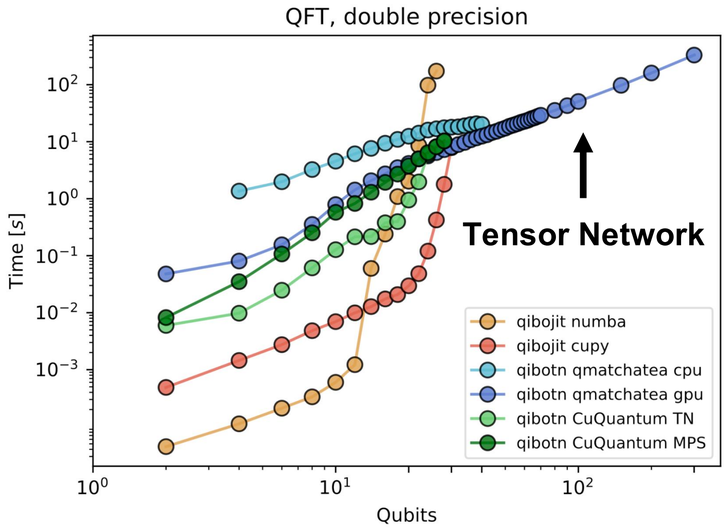
State-vector



Clifford



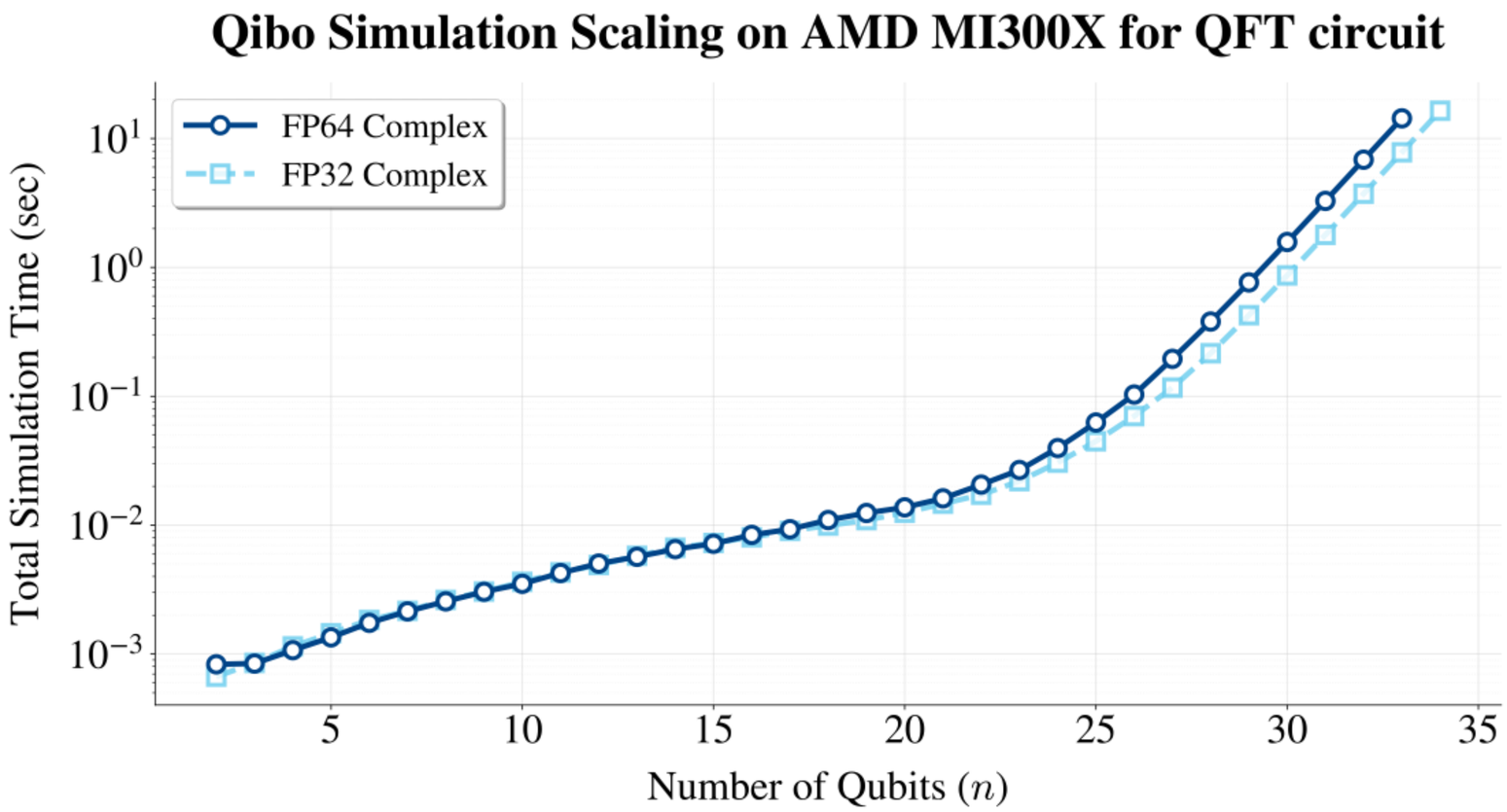
Tensor Network



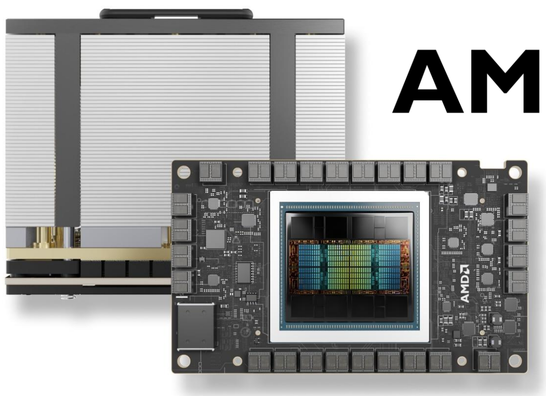
Quantum (6) (2022), arXiv:2009:01845

QIBO

Large-Memory GPU Support



State-vector simulation support for GPU models up to **MI300X 192GB**

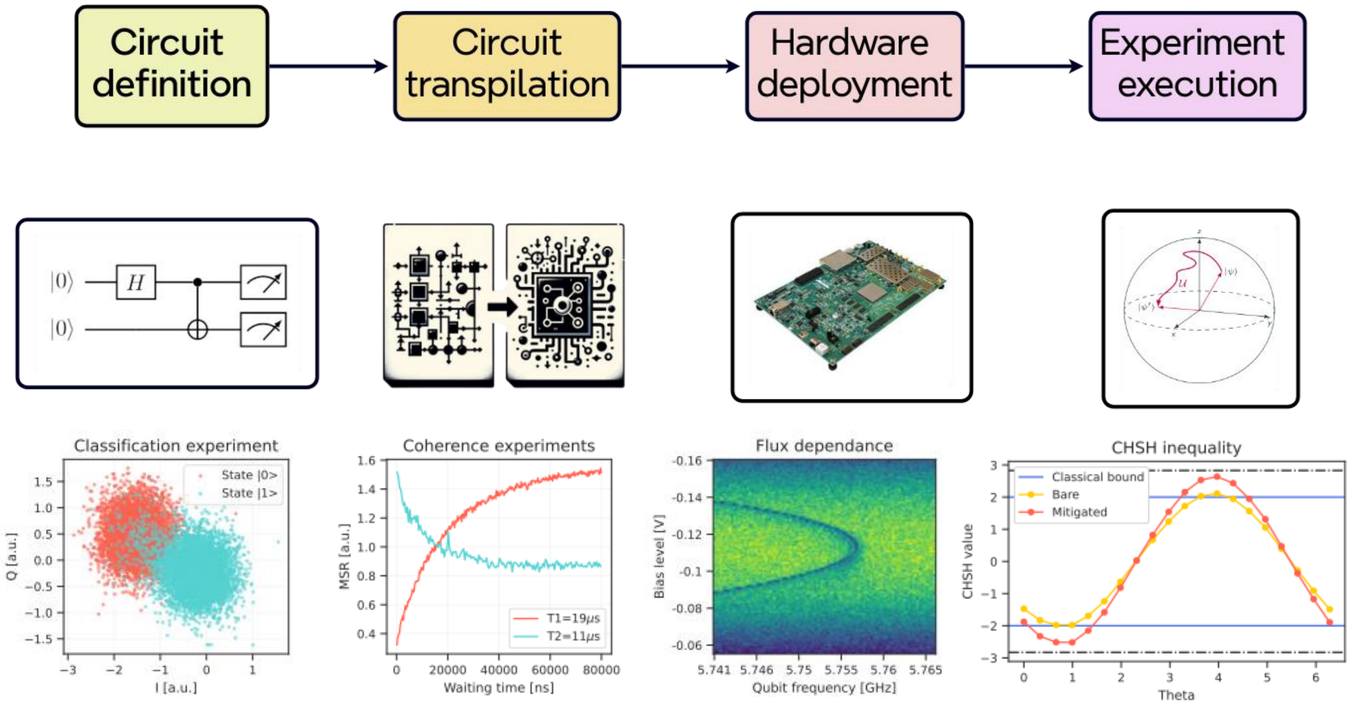


Quantum Control and Calibration

Qibolab is as a modular library for quantum control. Supports superconducting qubits. On-going progress for neutral atoms and trapped ions.

Qibocal is a software tool for characterization and calibration.

Qibosoq is a frontend for control operations based on QICK.



Quantum 8, 1247 (2024), arXiv:2308.06313, arXiv:2410.00101

Supported Hardware Electronics:

QBLOX

QUANTUM MACHINES

KEYSIGHT

AQSolotl

QICK

Supporting Partner for Qibosoq:

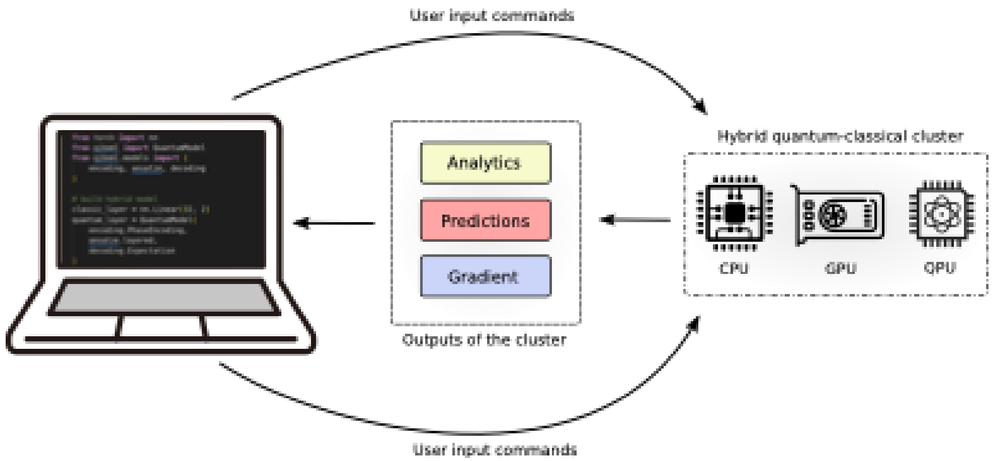
sdt

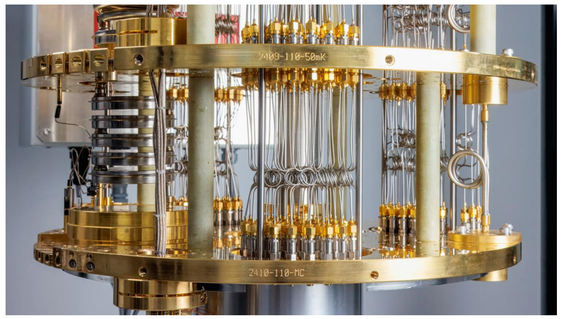
Multi-FPGA support

Application Examples

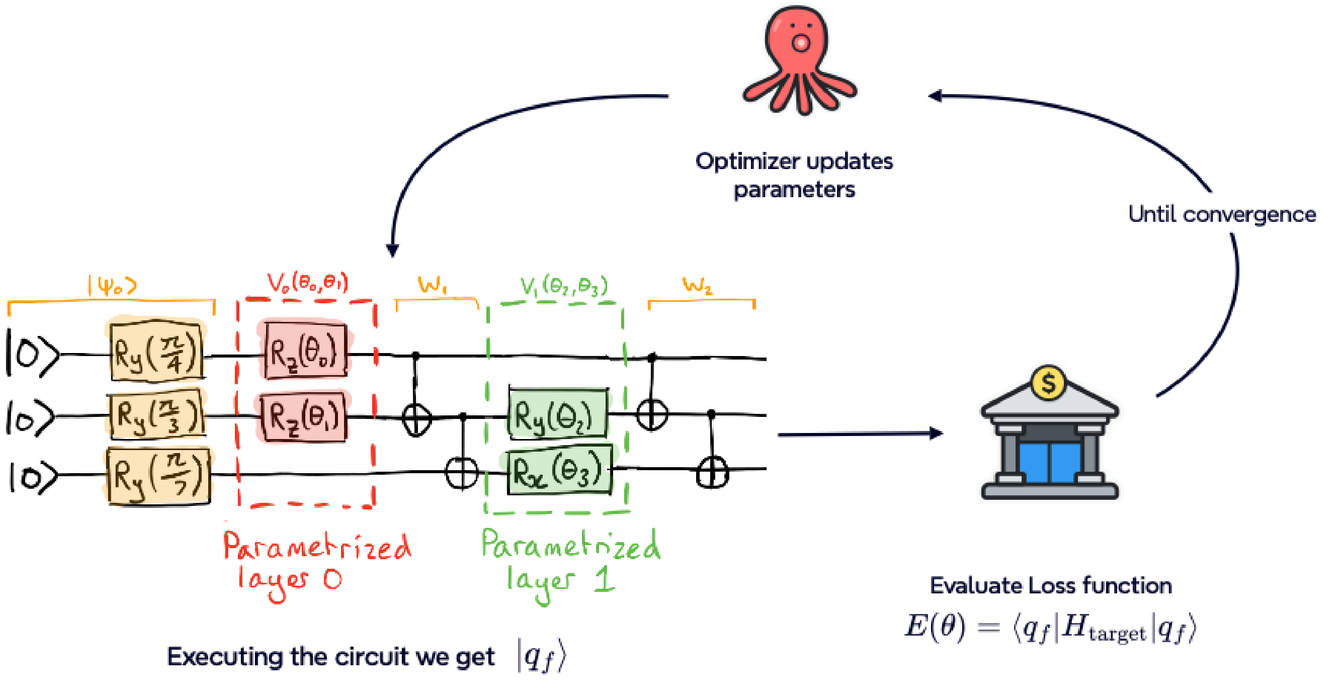
Quantum Machine Learning

Quantum Machine Learning involves quantum process units (QPU) into machine learning (ML) pipelines.





QML Pipeline



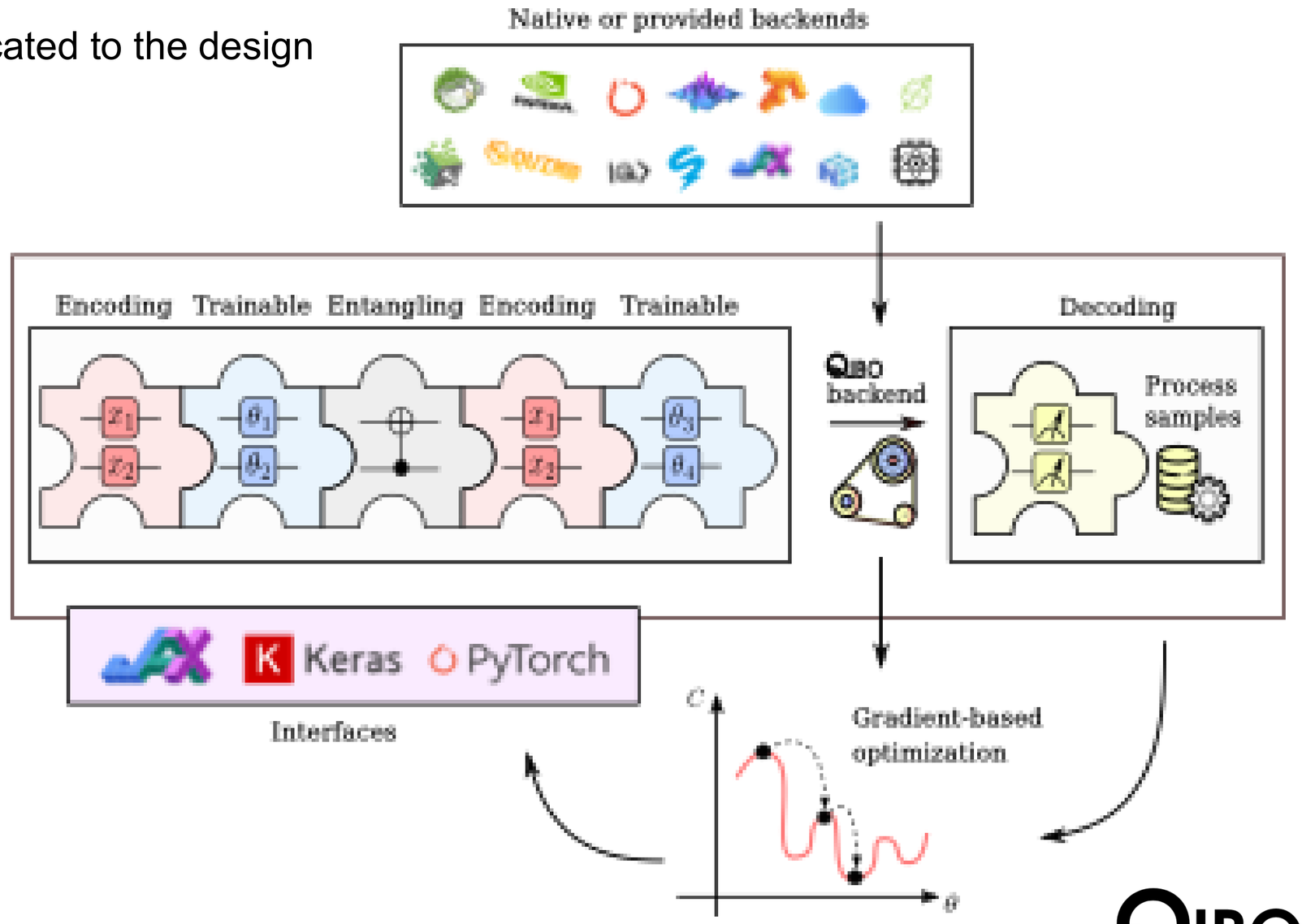
To fully realize the potential of a hybrid quantum-classical cluster, both classical and quantum components must be executable on CPUs, GPUs, and QPUs.

QiboML

Qibo also introduces a module dedicated to the design and training of QML models:

- Supervised learning
- Unsupervised learning
- Reinforcement learning
- VQC

QiboML:
ML frontend

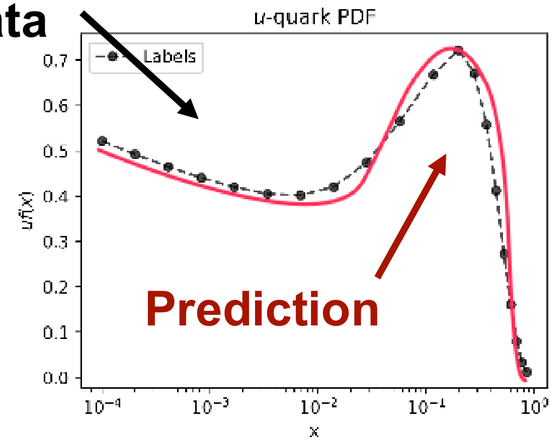


Example of Quantum Regression

M. Robbiati, A. Sopena, A. Papaluca, SC, arXiv:2311.05680

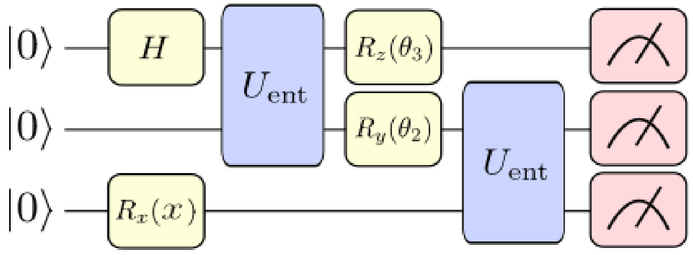
Classical Data

A target $f(x)$



Quantum Model

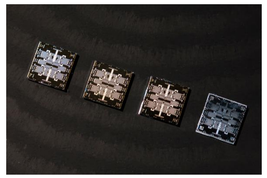
A parametric model $U(x;\theta)$



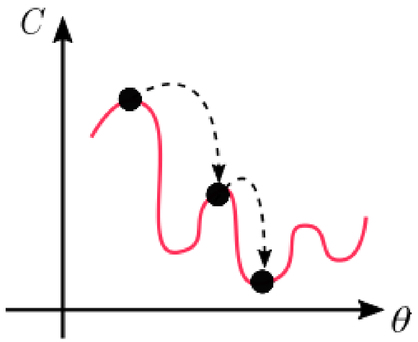
returning predictions

$$\tilde{f}(x;\theta) = \langle 0|U(x;\theta)^\dagger \hat{O} U(x;\theta)|0\rangle$$

Evaluated on QPU



Loss function (MSE)



A training algorithm

Classical Optimizer (i.e. Gradient Descent)

Machine Learning with QiboML: Universal Approximant

Setup:

- **Model executed:**
  Data reuploading on a single qubit to predict a target function.

- **Target function:**
  $$\sin^2 x - 0.3 \cos x$$

- **Scale of experiment:** 7000 circuits, 500 shots / circuit.

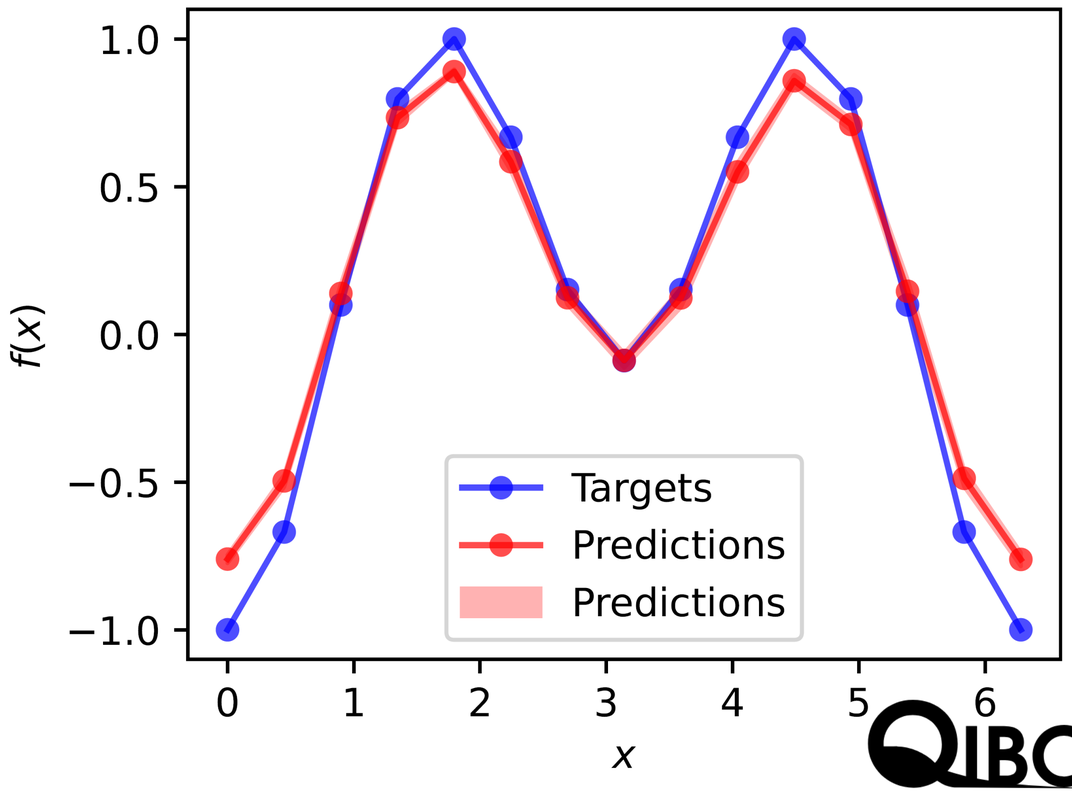
Implications:

- Small scale demonstration of QiboML library on Singapore's National Quantum Computing Hub (NQCH).

- Enables development of ML-based QEC.

Circuit Architecture

NQCH CQT NUS S14 Hardware



Outlook

Outlook

The present achievements:

- Qibo: a full-stack open-source framework for quantum computing.
- Institutional and community driven effort.
- Robust framework and coordination.

The future approach:

- Provide quantum computing integration facility with modular architecture and pulse-level access.
- Undertake core development work and partnership projects with industry.
- Scale up existing superconducting quantum computing architecture with short term future to integrate other qubit platforms.
- Further develop full-stack Qibo.
- Train talent for operation of quantum computing infrastructure.

Learn more:

https://qibo.science

QIBO

Thank you!